

Practicum 3

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1 Introduction

The bookwork problems were completed by hand and are included at the beginning of this document, followed by the rest of the questions accompanied by their code.

2 Bookwork

Practicum 3 Bookwork

Kieran Douglas

4.4

Model: $\log(\text{rent}) = \beta_0 + \beta_1 \log(\text{pop}) + \beta_2 \log(\text{avinc}) + \beta_3 \text{pctstu} + u$

i

Null: $\beta_3 = 0 \rightarrow H_0 \Leftrightarrow$ There is no CP effect of student pop. on rents.
 alt: $\beta_3 \neq 0 \rightarrow H_1 \Leftrightarrow$ There is an effect CP of student pop. on rents

ii I would expect positive signs for both β_1 and β_2 since generally in the U.S. places with higher populations are more expensive to live in due to increased housing demand. I would expect β_2 to also be positive because increases to average income generally go hand in hand with increased housing cost since landlords would be undercharging if people could or were willing to pay more. Additionally, higher income tends to prefer nicer rentals which may also increase cost. It makes sense that elasticity of rent wrt income and population would be generally increasing.

iii

estimated: $\ln(\text{rent}) = 0.43 + 0.66 \ln(\text{pop}) + 0.507 \ln(\text{avginc}) + 0.006 \text{pctstu}$

The statement is wrong because it reflects a misunderstanding of how log-log models should be interpreted. It may work for small percent changes in population, but since elasticities vary at different points on a curve we would expect it to be a bit different at 10% than at a 1% change. The correct interpretation uses the formula $\% \Delta y = ((1 + \% \Delta x)^{\beta} - 1) \times 100$ where $\% \Delta y = ((1 + 0.1)^{0.66} - 1) \times 100 = 0.631\%$ change

iv I will test the hypothesis $\{H_0: \beta_1 = 0, H_1: \beta_1 \neq 0\}$ using a two-sided t-test.

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} = \frac{0.066 - 0}{0.039} = 1.6923 \quad ; \quad 1\% \text{ critical value at } n-k-1$$

$\rightarrow 64 - 3 - 1 = 60$
 $\rightarrow t = 2.66$

I fail to reject the null and determine that there is insufficient evidence to suggest that $\beta_1 \neq 0$.

$$1.6923 < 2.66$$

4.8

model: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$, testing the null that $H_0: \beta_1 - 3\beta_2 = 1$

(i) Since $\text{Var}(\alpha X + \beta Y) = \alpha^2 \text{Var}(X) + \beta^2 \text{Var}(Y) + 2\alpha\beta \text{Cov}(X, Y)$

we can write $\text{Var}(\hat{\beta}_1 - 3\hat{\beta}_2) = 1^2 \text{Var}(\hat{\beta}_1) + 3^2 \text{Var}(\hat{\beta}_2) - 2 \cdot 3 \text{Cov}(\hat{\beta}_1, \hat{\beta}_2)$

$$\hookrightarrow \text{Var}(\hat{\beta}_1) + 9\text{Var}(\hat{\beta}_2) - 6\text{Cov}(\hat{\beta}_1, \hat{\beta}_2)$$

To complete the test we need the t -stat which requires SE

$$\text{SE}(\hat{\beta}_1 - 3\hat{\beta}_2) = \sqrt{\text{Var}(\hat{\beta}_1) + 9\text{Var}(\hat{\beta}_2) - 6\text{Cov}(\hat{\beta}_1, \hat{\beta}_2)}$$

(ii) $t = \frac{(\hat{\beta}_1 - 3\hat{\beta}_2) - 1}{\text{SE}(\hat{\beta}_1 - 3\hat{\beta}_2)}$ Since $t = \frac{\hat{\beta}_1 - \beta_{1,1}}{\text{SE}(\hat{\beta}_1)}$ and we test that $\beta_{1,1}$

we should reparameterize to $\hat{\theta} = \hat{\beta}_1 - 3\hat{\beta}_2$ so $t = \frac{\hat{\theta} - 1}{\text{SE}(\hat{\theta})}$

(iii) Reparameterizing $\theta = \beta_1 - 3\beta_2$ and $\hat{\theta} = \hat{\beta}_1 - 3\hat{\beta}_2$

We can write the regression equation

$$\hookrightarrow y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u \text{ as } y = \beta_0 + (\beta_1 - 3\beta_2)x_1 + \beta_2 x_2 + \beta_3 x_3 + u$$

$$\Rightarrow y = \beta_0 + \theta x_1 + \beta_2 (3x_1 + x_2) + \beta_3 x_3 + u \text{ if we write } \beta_1 = \theta + 3\beta_2$$

to obtain $\hat{\theta}$ and its SE, we need to construct a new variable

$$z = 3x_1 + x_2 \longrightarrow \hat{y} = \hat{\beta}_0 + \hat{\theta} x_1 + \hat{\beta}_2 z + \hat{\beta}_3 x_3 + u$$

and the SE for $\hat{\theta}$ would be $\text{SE}(\hat{\theta}) = \sqrt{\text{Var}(\hat{\beta}_1) + 9\text{Var}(\hat{\beta}_2) - 6\text{Cov}(\hat{\beta}_1, \hat{\beta}_2)}$

(5.4) $y = \beta_0 + \beta_1 x + u_i$ Under GM, OLS is BLUE. Under the GM's 1-4 we show that estimators of the form $\tilde{\beta}_1$ are consistent for the slope β_1 .

β_0 estimator $\tilde{\beta}_0 = \bar{y} - \tilde{\beta}_1 \bar{x}$ and $\tilde{\beta}_0 = \beta_0$

↳ This is true because the probability limit is $= \beta_0$:

$E(y_i) = \beta_0 + \beta_1 E(x_i) + E(u_i) \rightarrow$ Under GM, $E(u_i) = 0$ so...

↳ if $\mu_y = E(y_i) + \mu_x = E(x_i)$ then $\mu_y = \beta_0 + \beta_1 \mu_x \Rightarrow \beta_0 = \mu_y - \beta_1 \mu_x$

↳ LLN states that $\bar{y} \xrightarrow{P} \mu_y$ and $\bar{x} \xrightarrow{P} \mu_x$ so...

↳ since it's consistent for β_1 , $\tilde{\beta}_1 \xrightarrow{P} \beta_1$

↳ $\text{plim}(\tilde{\beta}_0) = \text{plim}(\bar{y} - \tilde{\beta}_1 \bar{x}) = \mu_y - \beta_1 \mu_x = \beta_0$

↳ so $\text{plim}(\tilde{\beta}_0) = \beta_0$ thus $\tilde{\beta}_0$ is consistent est. of β_0 .

(5.6) Given the equation $y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$ and $E(u|x) = 0$

↳ $E(x) = 0, E(x^2) = \text{Var}(x) = 1$
 $\& E(x^3) = 0$

(i) we can write $y = \alpha_0 + \beta_1 x + v$ because...

↳ $y = \beta_0 + \beta_1 x + \beta_2(x^2) + u \rightarrow y = \underbrace{(\beta_0 + \beta_2 E(x^2))}_{\alpha_0} + \beta_1 x + \underbrace{(\beta_2(x^2 - E(x^2)) + u)}_v$

↳ $\alpha_0 = \beta_0 + \beta_2 E(x^2)$ and $v = \beta_2(x^2 - E(x^2)) + u$

↳ thus $y = \alpha_0 + \beta_1 x + v$

(ii) $E(v|x)$ depends on x unless $\beta_2 = 0$ because...

↳ if $\beta_2 = 0$ then $v = u$ since $v = \beta_2(x^2 - E(x^2)) + u$

↳ otherwise $E(v|x)$ varies with x .

iii) $\text{Cov}(X, V) = 0$ because...

$$\hookrightarrow \text{Cov}(X, V) = E(XV) - E(X)E(V)$$

$$\hookrightarrow \text{since } E(V) = 0, \rightarrow \text{Cov}(X, V) = E(XV)$$

$$\hookrightarrow E(XV) = E[X\{\beta_2(X^2 - E(X^2)) + u\}] \rightarrow E[X(X^2 - E(X^2))]$$

$$\hookrightarrow \text{since } E(X) = 0, E(X^2) = 1, E(X^3) = 0 \Leftrightarrow 0 - 1 \cdot 0 = 0$$

$$\hookrightarrow \text{thus } E(XV) = 0 \text{ and } \text{Cov}(X, V) = 0 \text{ as well}$$

iv) I think that $\hat{\beta}_1$ is consistent for β_1 because the assumptions (minus $E(V|X) = 0$) imply that $\hat{\beta}_1 \xrightarrow{p} \beta_1$. Since $\text{Cov}(X, V) = 0$ and $\text{var}(X) > 0$, thus $\hat{\beta}_1$ is consistent for β_1 . I do think it would be biased though, as unbiasedness necessarily includes $E(V|X) = 0$ which we do not have here. The bias would however shrink asymptotically as n approaches ∞ .

v) Similar to the problem in Chapter 4, even if faced with a nonlinear true relationship we can gauge localized responses in y given some small change to X near its mean. Despite its biasedness, there can still be some utility in exploring these smaller localized changes in outcome.

vi) It is generally more valuable to consistently estimate β_1 and β_2 than just β_1 , since you can get a sense of the complete slope and quadratic relationship at play. The marginal effect $\beta_1 + 2\beta_2 X$ can then be computed at any point, not just at or around the mean. This can drastically improve both the quality and scope of an analysis.

6.4

Model estimated: $\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{educ} \cdot \text{pareduc} + \beta_3 \text{exper} + \beta_4 \text{tenure} + u$

i) In decimal form the return to another year of education in this model is $\Delta \log(\text{wage}) / \Delta \text{educ} = \beta_1 + \beta_2 \text{pareduc}$

$\rightarrow \frac{\Delta \log(\text{wage})}{\Delta \text{educ}} = \beta_1 + \beta_2 \text{pareduc} \rightarrow$ another year of educ returns $a(\beta_1 + \beta_2 \text{pareduc}) 100\%$ Change in wage where absent the $\times 100$ it is a proportion.

I would expect β_2 to be positive, as parents who are more educated probably have children who are more likely to be educated, and both are probably more likely to see higher earnings than those with less education.

ii) The coefficient on the interaction term states that the returns to education are $(0.078\% \times \text{years of parental education} \times 4.7\%)$ % higher for each additional year that a parent was educated. In other words, the marginal effect of an additional year of education increases by 0.078% for each additional year of parental education.

iii) When pareduc is added, the interpretation of the interaction coefficient changes slightly. It now shows the marginal effect of schooling on wages decreases by 0.16% per additional year of parental education. The estimated return to education depends negatively on parental education according to this model, but honestly the true effects are now a bit unclear.

Test:

H_0 : return to education does not depend on parental education: $\beta = 0$
 H_1 : return to education does depend on parental education: $\beta \neq 0$

$$t = \frac{\beta_1 - 0}{SE(\beta_1)} \Rightarrow \frac{0.0016}{0.0012} = -1.33 \rightarrow \text{with } n = 722, \text{crit: } \pm 1.92$$

$\rightarrow -1.33 > -1.92$ so fail to reject the null.

3 Code

Practicum 3

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2. Replicating Nerlove's Classic Results on Scale Economies

A.

The purpose of this assignment is to replicate some of the principles of returns-to-scale reported by Nerlove in his 1955 article. His estimated equation was as follows:

$$\ln C^* = \beta_o + \beta_y \ln(y) + \beta_1 \ln p_1^* + \beta_2 \ln p_2^*$$

For part A I will generate the variables necessary for estimating his parameters. includes $\ln CP3 = \ln(COSTS/PF)$, $\ln P13 = \ln(PL/PF)$, $\ln P23 = \ln(PK/PF)$, $\ln KWH = \ln(KWH)$

I will do this using the mutate function in Tidyverse.

```
# Load packages and data from the NERLOV file
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.1      v tibble     3.2.1
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.0.2

-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```

library(readxl)

nerlov <- read_excel("/Users/kieran/Documents/MASTERS/METRICS/code/metrics/practicum_3_files,

# Clean data and generate new variables per assignment request

clean <- nerlov |>
  mutate(
    LNCP3 = log(COSTS/PF),
    LNP13 = log(PL/PF),
    LNP23 = log(PK/PF),
    LNKWH = log(KWH)
  )

# Preview order of LNKWH

print(clean$LNKWH)

```

```

[1] 0.6931472 1.0986123 1.3862944 1.3862944 1.6094379 2.1972246 2.3978953
[8] 2.5649494 2.5649494 3.0910425 3.2188758 3.2188758 3.5553481 3.6635616
[15] 3.7612001 4.1431347 4.2195077 4.3944492 4.4308168 4.2904594 4.5951199
[22] 4.6151205 4.7791235 4.7874917 4.8040210 4.8675345 4.9272537 5.0039463
[29] 5.2781147 5.2832037 5.3423343 5.3659760 5.3936275 5.4553211 5.4595855
[36] 5.5333895 5.6312118 5.6698809 5.6698809 5.6869754 5.7004436 5.7807435
[43] 5.8081425 5.8230459 5.8664681 5.8664681 6.0306853 6.0402547 6.1224928
[50] 6.1820849 6.2461068 6.3099183 6.3332796 6.3385941 6.3835066 6.5087691
[57] 6.5453497 6.5778614 6.6093492 6.6783421 6.6846117 6.6945621 6.6982681
[64] 6.7511015 6.7569324 6.8123451 6.8167359 6.8287121 6.8916259 6.8987145
[71] 6.9077553 7.0012456 7.0112140 7.0192967 7.0228681 7.0361485 7.0527210
[78] 7.0613344 7.0647590 7.1024994 7.1538338 7.1631724 7.1623975 7.1936858
[85] 7.2247534 7.2584122 7.2957351 7.3112184 7.3427792 7.4079243 7.4193806
[92] 7.4854916 7.5126175 7.5137092 7.5164333 7.4882935 7.5590383 7.5652753
[99] 7.6148054 7.6290039 7.6420444 7.7079615 7.7424020 7.7583335 7.7634464
[106] 7.7693786 7.8042514 7.8066964 7.8268421 7.8359746 7.8539931 7.8659554
[113] 7.9620673 8.0040315 8.0715309 8.0974263 8.1053075 8.1599467 8.1713169
[120] 8.2411762 8.2534880 8.2975435 8.3468793 8.3675324 8.4104985 8.4688429
[127] 8.5711130 8.5722494 8.6425916 8.6448826 8.6688837 8.6995147 8.7191540
[134] 8.7219283 8.8808636 8.9728443 9.0382463 9.0643893 9.0810286 9.1573614
[141] 9.2059307 9.3481003 9.3755158 9.5721322 9.7243011

```

```
# The observations are by size of output, looks good!
```

B.

Now that I have all of the data in line, I will estimate the following model:

$$\ln C^* = \beta_o + \beta_y \ln(y) + \beta_1 \ln p_1^* + \beta_2 \ln p_2^*$$

Running this model I find coefficient estimates that slightly differ from those found by Nerlove, with β_y , β_1 and β_2 estimates coming in at 0.7207, 0.5929, and -0.0074 , respectively, with standard errors reporting as 0.0174, 0.2046, and 0.1907.

```
model1 <- lm(data = clean, LNCP3~LNKWH+LNP13+LNP23)
summary(model1)
```

Call:

```
lm(formula = LNCP3 ~ LNKWH + LNP13 + LNP23, data = clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.01200	-0.21759	-0.00752	0.16048	1.81922

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-4.690789	0.884871	-5.301	4.34e-07	***
LNKWH	0.720688	0.017436	41.334	< 2e-16	***
LNP13	0.592910	0.204572	2.898	0.00435	**
LNP23	-0.007381	0.190736	-0.039	0.96919	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3918 on 141 degrees of freedom

Multiple R-squared: 0.9316, Adjusted R-squared: 0.9301

F-statistic: 640 on 3 and 141 DF, p-value: < 2.2e-16

C.

I will now construct a confidence interval for β_y . This ends up being (0.67516030.7662147), which means that using the same procedure on repeated samples, we would expect 99% of intervals to contain the true population coefficient. Based on this, we could determine that at

the 99% confidence interval, the null hypothesis that $\beta_y = 1$ is rejected, meaning that with a high degree of certainty we could say that values of 1 are not expected as the true population coefficient under this model. This implies that we would reject the null hypothesis that returns to scale were constant, because we can say with a high degree of certainty that they will be increasing since our confidence interval shows that we would expect 99% of intervals (0.68, 0.77) to contain the true population coefficient. If the interval contained 1 we may say that it is plausible for the model to display constant returns to scale, but since it is <1 , we can say with a reasonable degree of confidence that in most all cases a 1% increase in the output will raise cost by less than 1% which implies increasing returns to scale. Computing the point estimate for returns to scale, I find an r of 1.388. Since this is greater than 1, we again confirm observation of increasing returns to scale. This means that if we were to double inputs, outputs would more than double! Since returns to scale are increasing, in this case we have positive economies of scale. This means that as output increases, cost increases at a decreasing rate, or average cost falls.

```
# Construct a 99% confidence interval using the estimated model
# for the coefficient on LNKWH ($beta_y$)
confint(model1, "LNKWH", level = 0.99)
```

```
          0.5 %      99.5 %
LNKWH 0.6751603 0.7662147
```

```
# compute the point estimate of returns to scale r,
# where  $r=1/b_y$ . It is 1.388
1/0.720688
```

```
[1] 1.387563
```

D.

To calculate the implied estimate of α_2 from part C, I have to multiply the coefficient β_2 by the point estimate for returns to scale, r . This yields $-0.007381 * 1.388 = -0.01024$. This estimate is very close to 0, which makes me think that Nerlove was dissatisfied with his estimate of α_2 since it would imply that the second input (capital) was either not effective or negatively productive which wouldn't really make sense for a necessary input. Additionally, since the estimate is so close to 0, it seems likely that a confidence interval would contain 0 deeming it non statistically significant which also wouldn't make sense. This finding contradicts what we would expect of Cobb-Douglas, indicating that something weird might be going on with the data we have that is causing it to inaccurately reflect the true production process.

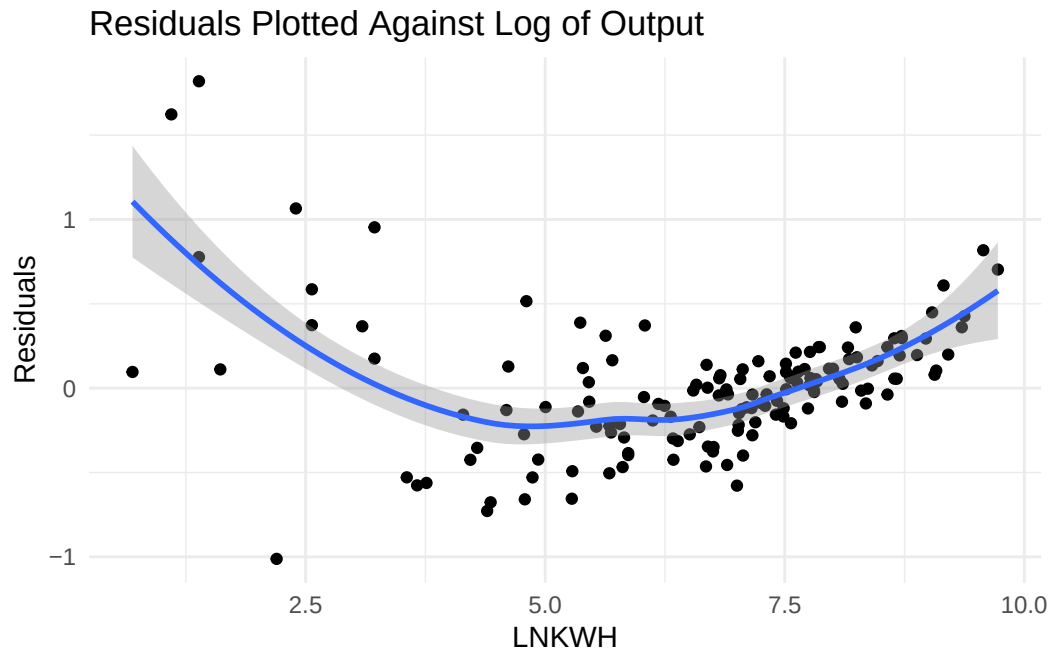
E.

I also find a U-shaped pattern when plotting residuals against LNKWH. I think that this could be a sign of model misspecification, where the model we are estimating does not adequately capture the underlying relationship in the data. The implication here is that the true relationship between output and cost is likely nonlinear in some way, and our linearity assumption is failing. Next, calculating the correlation between the residuals and LNKWH I find a coefficient of effectively 0 ($-9.565708 \times 10^{-17}$). This is not surprising as OLS ensures that residuals are not correlated with regressors in the model by optimizing the fit such that linear association between residuals and regressors is removed. This however is not indicative of a perfect model, and provides a good example of why it's important to check things graphically to ensure that we are getting what we expected.

```
# create a new row with residuals from the model
clean <- clean |>
  mutate(
    resid = residuals(model1)
  )

# plot against log of output
ggplot(data = clean, mapping = aes(y = resid, x = LNKWH)) +
  geom_point() +
  geom_smooth(method = loess) +
  theme_minimal() +
  labs(title = "Residuals Plotted Against Log of Output", x = "LNKWH", y = "Residuals")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
# compute sample correlation of residuals with LNKWH across the sample
cor(clean$LNKWH, clean$resid)
```

```
[1] -9.565708e-17
```

3. Assessing Alternative Returns-to-Scale Specifications

A.

My attempted replication of Nerlove's results was actually far closer than I was expecting. Both the estimates and the standard errors were within a very small distance of each other, and any spread appeared to be relatively unsystematic, varying from being greater than or less than with within and between subsets. I would estimate that any observed discrepancies are probably due to the fact that I used natural logarithms in my transformations while in the reported data from Nerlove common logarithms were used.

```
# generate subsamples of data for each set of 29 rows
ss1 <- clean |>
  filter(
    ORDER <200
  )
```

```

ss2 <- clean |>
  filter(
    ORDER > 200,
    ORDER < 300
  )
ss3 <- clean |>
  filter(
    ORDER > 300,
    ORDER < 400
  )
ss4 <- clean |>
  filter(
    ORDER >400,
    ORDER <500
  )
ss5 <- clean |>
  filter(
    ORDER > 500
  )
# now I will estimate a model for each of these subsets
# and extract the coefficients to combine in a table
library(broom)

ss1_model <- lm(data = ss1, LNCP3~LNKWH+LNP13+LNP23)
ss2_model <- lm(data = ss2, LNCP3~LNKWH+LNP13+LNP23)
ss3_model <- lm(data = ss3, LNCP3~LNKWH+LNP13+LNP23)
ss4_model <- lm(data = ss4, LNCP3~LNKWH+LNP13+LNP23)
ss5_model <- lm(data = ss5, LNCP3~LNKWH+LNP13+LNP23)

# create a combined coefficients vector to use in table
coef_combined <- list(ss1_model, ss2_model, ss3_model, ss4_model, ss5_model)
# create and print tables for each subset
coef_tbl <- lapply(seq_along(coef_combined), function(i) {
  tidy(coef_combined[[i]]) %>%
    mutate(coef_combined = paste0("model", i))
})
combined <- bind_rows(coef_tbl)
print(coef_tbl)

```

```
[[1]]
```

```
# A tibble: 4 x 6
```

```
term      estimate std.error statistic  p.value coef_combined
```

	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	(Intercept)	-3.34	3.15	-1.06	0.298	model1
2	LNKWH	0.400	0.0845	4.74	0.0000731	model1
3	LNP13	0.615	0.729	0.843	0.407	model1
4	LNP23	-0.0814	0.706	-0.115	0.909	model1

[[2]]

A tibble: 4 x 6

	term	estimate	std.error	statistic	p.value	coef_combined
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	(Intercept)	-6.49	1.41	-4.59	0.000107	model2
2	LNKWH	0.658	0.116	5.66	0.00000684	model2
3	LNP13	0.0938	0.274	0.342	0.735	model2
4	LNP23	0.378	0.277	1.37	0.184	model2

[[3]]

A tibble: 4 x 6

	term	estimate	std.error	statistic	p.value	coef_combined
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	(Intercept)	-7.33	1.69	-4.34	0.000205	model3
2	LNKWH	0.938	0.198	4.74	0.0000732	model3
3	LNP13	0.402	0.199	2.02	0.0546	model3
4	LNP23	0.250	0.187	1.34	0.193	model3

[[4]]

A tibble: 4 x 6

	term	estimate	std.error	statistic	p.value	coef_combined
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	(Intercept)	-6.55	1.16	-5.62	0.00000757	model4
2	LNKWH	0.912	0.107	8.48	0.00000000791	model4
3	LNP13	0.507	0.187	2.70	0.0121	model4
4	LNP23	0.0934	0.164	0.569	0.575	model4

[[5]]

A tibble: 4 x 6

	term	estimate	std.error	statistic	p.value	coef_combined
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	(Intercept)	-6.71	1.05	-6.42	1.02e- 6	model5
2	LNKWH	1.04	0.0650	16.1	1.09e-14	model5
3	LNP13	0.603	0.197	3.05	5.30e- 3	model5
4	LNP23	-0.289	0.175	-1.66	1.10e- 1	model5

B.

Using the subset models I ran in the previous section, I compute point estimates of returns to scale ($r=1/b_y$) for each of the five subsamples. For subsets 1-5 I find point estimates of returns to scale of 2.500, 1.519, 1.066, 1.096, and 0.961 respectively. These point estimates suggest that as output increases, cost generally decreases suggesting on average positive returns to scale. There are discrepancies that I observe though, as the magnitude at which the returns to scale are increasing are decreasing as output increases. Additionally, in the final subset, we see a switch to what may be either slightly decreasing or constant returns to scale. This indicates that at larger quantities bottlenecks or constraints may be experienced that limit continued gains. I think this suggests an alternative specification that is perhaps broken up by quantity and interpreted in terms of the subsections rather than their aggregate. The implications are that returns to scale may not be consistent in this case (or any really), reflecting the U-shaped residuals plot we observed in the previous section.

```
# subsample 1  
1/0.4
```

```
[1] 2.5
```

```
#subsample 2  
1/0.658
```

```
[1] 1.519757
```

```
# subsample 3  
1/0.938
```

```
[1] 1.066098
```

```
# subsample 4  
1/0.912
```

```
[1] 1.096491
```

```
#subsample 5  
1/1.04
```

```
[1] 0.9615385
```

C.

I think that since the β_y coefficients vary pretty substantially across groups but the β_1 and β_2 coefficients do not as much. Basically, the output causes pretty different cost responses but the responsiveness to different input prices remains relatively stable. Given this, I think that by allowing intercept terms and β_y coefficients to vary while holding the others constant across groups. This should allow for us to better control for the observed heterogeneity, generating a model that is a better specification of our sample given its observed shape in earlier parts. After replicating Nerlove's model by including all of the dummies and their interactions with β_y I was able to achieve almost a perfect copy of his estimates. Comparing mine to Nerlove's, the ones that are the most different are our estimated coefficients on the interaction between the second dummy (those between 200 and 300) in which he estimated 0.651 while I estimated 0.648, and our R^2 which he estimated as 0.95 while I found it to be 0.96. As you can see, these are VERY close to one another and any differences probably have to do again with the fact that I used natural logarithms in transformation while Nerlove used common logarithms. I believe that this also explains why all of my standard errors are slightly smaller than those reported by Nerlove.

```
# I am going to start by creating dummy variables representing each group,
# followed by variables that multiply that dummy by LNKWH to act
# as proper interaction terms and finally rerun the regression
# including our new variables.
# the dummies are called d1-d5
clean <- clean |>
  mutate(
    d1 = ORDER <200,
    d2 = ORDER >=200 & ORDER <300,
    d3 = ORDER >=300 & ORDER <400,
    d4 = ORDER >=400 & ORDER <500,
    d5 = ORDER >=500,
    d1_LNKWH = d1*LNKWH,
    d2_LNKWH = d2*LNKWH,
    d3_LNKWH = d3*LNKWH,
    d4_LNKWH = d4*LNKWH,
    d5_LNKWH = d5*LNKWH
  )
# rerun the model including the dummy variables
model2 <- lm(data = clean, LNCP3~d1+d2+d3+d4+d5+d1_LNKWH+d2_LNKWH+d3_LNKWH+d4_LNKWH+d5_LNKWH)
summary(model2)
```

Call:

```
lm(formula = LNCP3 ~ d1 + d2 + d3 + d4 + d5 + d1_LNKWH + d2_LNKWH +
    d3_LNKWH + d4_LNKWH + d5_LNKWH + LNP13 + LNP23, data = clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.46008	-0.14123	0.00942	0.13061	1.10602

Coefficients: (1 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-8.08336	1.38018	-5.857	3.51e-08 ***
d1TRUE	3.90353	1.16071	3.363	0.00101 **
d2TRUE	3.03092	1.43979	2.105	0.03716 *
d3TRUE	1.45324	2.36033	0.616	0.53915
d4TRUE	1.35475	2.39642	0.565	0.57281
d5TRUE	NA	NA	NA	NA
d1_LNKWH	0.39688	0.04307	9.214	6.04e-16 ***
d2_LNKWH	0.64816	0.14724	4.402	2.18e-05 ***
d3_LNKWH	0.88478	0.29727	2.976	0.00347 **
d4_LNKWH	0.90874	0.27365	3.321	0.00116 **
d5_LNKWH	1.06274	0.13134	8.091	3.26e-13 ***
LNP13	0.42561	0.16317	2.608	0.01014 *
LNP23	0.10373	0.15221	0.681	0.49675

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3075 on 133 degrees of freedom

Multiple R-squared: 0.9602, Adjusted R-squared: 0.9569

F-statistic: 292 on 11 and 133 DF, p-value: < 2.2e-16

D.

Using the model I estimated in the previous section, I will now use the point estimate returns to scale formula from part B. to compute the implied estimate of returns to scale for each of the five subsamples. From subsamples 1-5 I find point estimate returns to scale of 2.52, 1.54, 1.13, 1.10, and 0.94 respectively. Similar to my findings in part B. we observe increasing returns to scale that decrease with each subsequent output increase. This trend continues until we reach subsample 5 which exhibits decreasing (perhaps constant) returns to scale, which seems to be a common trend as output scales up due to physical constraints. Basically, my estimated scale economies are quite strong when output is very small, but as output grows the benefits diminish. Once we reach the highest reported output point we even notice constant or decreasing average cost per unit meaning that further growth is no longer bringing an advantage as it once did.

```
# subsample 1  
1/0.39688
```

```
[1] 2.519653
```

```
#subsample 2  
1/0.64816
```

```
[1] 1.542829
```

```
# subsample 3  
1/0.88479
```

```
[1] 1.130212
```

```
# subsample 4  
1/0.90874
```

```
[1] 1.100425
```

```
#subsample 5  
1/1.06274
```

```
[1] 0.9409639
```

E.

I will now compute an F-Test to determine whether the constrained model used in part A. fits the data better or worse than the unconstrained model used in part C. Here, my null hypothesis H_0 is that the constraints in part A. create a model that sufficiently explains how cost responds to quantity, while my alternative hypothesis H_1 is that the relaxed model specified in part C. better fits the data in explaining how cost responds to quantity. The F-test in the case of restricted versus unrestricted models comes in the form of $((SSR_{restricted} - SSR_{unrestricted})/q)/(SSR_{unrestricted}/(n - k_{unrestricted}))$ where n is the number of observations and k is the number of parameters in the unrestricted model. I will compute this using an ANOVA test in R. From the test, we see that the hypothesis that allowing each group to have its own intercept significantly improves the model fit, rejecting the null. The significantly smaller SSR found in the unrestricted model and the F-stat and its corresponding p-value

being VERY close to zero gives my confidence that the unrestricted model does a better job at explaining the sample data. The findings suggest and support the idea that each different group has different intercepts and output elasticities meaning that the effect of output on cost varies by group significantly.

```
# run anova between restricted and unrestricted models
anova(model1, model2)
```

Analysis of Variance Table

```
Model 1: LNCP3 ~ LNKWH + LNP13 + LNP23
Model 2: LNCP3 ~ d1 + d2 + d3 + d4 + d5 + d1_LNKWH + d2_LNKWH + d3_LNKWH +
          d4_LNKWH + d5_LNKWH + LNP13 + LNP23
   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1      141 21.640
2      133 12.577  8      9.063 11.98 8.901e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

F.

Running this model gives me estimates that are also quite similar to those of Nerlove. The small differences that exist are likely due to the difference in logs used as noted in previous sections. Similarly, my R^2 is pretty spot on, less than 0.01 away from Nerlove's. Next, I conduct a joint hypothesis test to determine whether returns to scale are constant using a 95% confidence interval. My first null hypothesis holds that returns to scale are nonconstant, that is $\beta_y/ = 1$ and $\beta_{yy}/ = 0$, while my other null hypothesis holds that returns to scale are constant, or $\beta_y = 1$ and $\beta_{yy} = 0$. The joint null hypothesis test yields a massive F-statistic of 252.47 and a p-value that is far below our threshold, meaning we can strongly reject the null hypothesis of constant returns to scale. I think that here the joint hypothesis test gives me a statistical test with greater validity since it evaluates the combined effects of relaxing both restrictions at once which is useful here since the parameters are correlated. The individual t-tests evaluate each parameter separately which could ignore estimate covariance and lead to inaccurate findings. Finally, since the returns to scale in the expanded model vary with the level of output, I compute the implied range of returns to scale estimates using the median value of LNY in each of the subsamples. Computing the r values I find an implied returns of scale for subsets 1-5 of 1.8778161, 1.3498233, 1.1616076, 1.0738059, and 0.9724344 respectively. This reflects earlier findings, though at a slightly lower magnitude towards the higher end, of positive returns to scale diminishing as quantity increases up to the point where it is slightly below constant returns to scale suggesting that further growth in output is no longer bringing the same gains it once did at lower quantities.


```
# run model3 with the new term
model3 <- lm(LNCP3 ~ LNKWH + I(LNKWH^2) + LNP13 + LNP23, data = clean)
summary(model3)
```

Call:

```
lm(formula = LNCP3 ~ LNKWH + I(LNKWH^2) + LNP13 + LNP23, data = clean)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.38249	-0.13715	0.00814	0.12782	1.13544

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3.764648	0.701727	-5.365	3.27e-07 ***
LNKWH	0.152547	0.061860	2.466	0.01487 *
I(LNKWH^2)	0.050514	0.005364	9.418	< 2e-16 ***
LNP13	0.480586	0.161072	2.984	0.00336 **
LNP23	0.074166	0.150016	0.494	0.62181

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3076 on 140 degrees of freedom

Multiple R-squared: 0.9581, Adjusted R-squared: 0.9569

F-statistic: 800.7 on 4 and 140 DF, p-value: < 2.2e-16

```
# run a joint null hypothesis test
library(car)
```

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

recode

The following object is masked from 'package:purrr':

some

```
linearHypothesis(model3, c("LNKWH = 1", "I(LNKWH^2) = 0"))
```

Linear hypothesis test:

LNKWH = 1

I(LNKWH^2) = 0

Model 1: restricted model

Model 2: LNCP3 ~ LNKWH + I(LNKWH^2) + LNP13 + LNP23

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	142	61.027				
2	140	13.248	2	47.779	252.47	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# calculate values of returns to scale for all 5 subsamples
```

```
# to start I find meadian for LNY for each subsample
```

```
medians <- c(
  median(ss1$LNKWH, na.rm = TRUE),
  median(ss2$LNKWH, na.rm = TRUE),
  median(ss3$LNKWH, na.rm = TRUE),
  median(ss4$LNKWH, na.rm = TRUE),
  median(ss5$LNKWH, na.rm = TRUE)
)
print(medians)
```

```
[1] 3.761200 5.823046 7.011214 7.707962 8.668884
```

```
# now that I have my medians I will plug them into the r equation
```

```
r_vals <- c(
  1/(0.152547 + 2 * 0.050514 * 3.761200),
  1/(0.152547 + 2 * 0.050514 * 5.823046),
  1/(0.152547 + 2 * 0.050514 * 7.011214),
  1/(0.152547 + 2 * 0.050514 * 7.707962),
  1/(0.152547 + 2 * 0.050514 * 8.668884)
)
print(r_vals)
```

```
[1] 1.8778161 1.3498233 1.1616076 1.0738059 0.9724344
```

4. Comparing Returns-to-Scale Estimates from 1955 with Updated 1970 Data

A.

To start my analysis, I will construct the necessary variables to estimate Nerlove's equation by OLS. I generate the variables LNC70, LNY70, LNP170, and LNP270. Next I find that the sample mean for KWH70 is 8999.727 while that of KWH is just 2133.083, a pretty substantial difference. This indicates that on average firms are generating significantly larger amounts of energy in 1970 than they were in 1955. Based on this information, we may expect much smaller returns to scale, potentially even decreasing returns to scale. Our previous model showed that as output scaled up the returns to scale diminished. Since we see such a jump in KWH produced between then and the 1970 data, we may expect to see that trend embodied. There is also the possibility that technological improvements and expansion of productive capacity could offset this expectation however, and we could see similar if not greater returns to scale in 1970 than we did before.

```
# load new update data
update <- read_excel('/Users/kieran/Documents/MASTERS/METRICS/code/metrics/practicum_3_files')
# clean data and add new generated variables
update_clean <- update |>
  mutate(
    LNC70 = log(COST70/PF70),
    LNY70 = log(KWH70),
    LNP170 = log(PL70/PF70),
    LNP270 = log(PK70/PF70)
  )
# compute sample means of KWH to compare
mean(update_clean$KWH70)
```

```
[1] 8999.727
```

```
mean(clean$KWH)
```

```
[1] 2133.083
```

B.

I estimate the model $LNC70 = \beta_0 + \beta_1 LNY70 + \beta_2 LNP170 + \beta_3 LNP270 + u$ and find a 99% confidence interval which comes out to (0.7923, 0.8584). Based on these results, I can

determine that the null hypothesis that $\beta_y = 1$ (indicating constant returns to scale) can be rejected because the 99% confidence interval is well below 1. The difference between our highest CI boundary is more than a couple standard errors away from 1, giving me strong confidence that given this model we can reject the null. Additionally, we can see that since β_y is well below 1, indicating increasing returns to scale. This is backed by the r value of 1.212 which indicates that as quantity increases, we get more bang for our buck through efficiency gains. Compared to the findings in section 2, this is a bit surprising as I find a larger coefficient on β_y which according to previous results shouldn't be the case since quantity increased so much.

```
# run model
model4 <- lm(data = update_clean, LNC70~LNY70+LNP170+LNP270)
summary(model4)
```

Call:

```
lm(formula = LNC70 ~ LNY70 + LNP170 + LNP270, data = update_clean)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.49086	-0.13036	-0.01245	0.08125	0.86443

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-8.09237	0.56393	-14.350	<2e-16 ***
LNY70	0.82533	0.01257	65.677	<2e-16 ***
LNP170	0.14521	0.10903	1.332	0.1861
LNP270	0.16496	0.09803	1.683	0.0957 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.211 on 95 degrees of freedom

Multiple R-squared: 0.9785, Adjusted R-squared: 0.9778

F-statistic: 1441 on 3 and 95 DF, p-value: < 2.2e-16

```
# construct confidence interval
confint(model4, "LNY70", level = 0.99)
```

	0.5 %	99.5 %
LNY70	0.7923018	0.8583663

```
# calculate the r value
1/0.82533
```

```
[1] 1.211637
```

C.

Now I add a squared term and run a more generalized version of the previous model to account for potential nonlinearity, a culprit observed in the 1950's data. This model outputs coefficients for $\beta_y, \beta_{yy}, \beta_1, \text{ and } \beta_2$ of 0.3014, 0.0367, 0.2941, and 0.0507 respectively and an R^2 of 0.99. Next I run a hypothesis test on the null that returns to scale do not vary with output, or that $\beta_{yy} = 0$ using a 95% confidence interval. To do this I compare the t statistic reported in the regression to the respective p value for the coefficient on $I(LNY70^2)$ which, since the p value is VERY close to 0, allows us to reject the null and determine that there is sufficient evidence to suggest that returns to scale vary with output level. Next I test the joint null hypothesis that returns to scale are constant, that is $\beta_{yy} = 0$ and $\beta_2 = 1$. I find a significantly smaller RSS than in the constricted model, a massive f statistic and a p value that is effectively 0, allowing me to reject the null and determine that the unrestricted specification is a better fit for the sample. This means that there is strong evidence that the relationship between cost and output is nonlinear. These findings are not mutually exclusive, and in fact complement each other nicely. Both the individual t-test and the joint f-test yield results suggesting that the underlying relationship involves some nonlinear element, and that perhaps the restricted specification is inadequate.

```
# run the updated model
model5 <- lm(data = update_clean, LNC70~LNY70+I(LNY70^2)+LNP170+LNP270)
summary(model5)
```

Call:

```
lm(formula = LNC70 ~ LNY70 + I(LNY70^2) + LNP170 + LNP270, data = update_clean)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.36949	-0.09177	0.00485	0.07526	0.33890

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-7.10760	0.37778	-18.814	< 2e-16 ***
LNY70	0.30138	0.04678	6.443	4.97e-09 ***

```

I(LNY70^2)    0.03675    0.00323   11.376   < 2e-16 ***
LNP170        0.29408    0.07229    4.068  9.86e-05 ***
LNP270        0.05073    0.06471    0.784    0.435

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1376 on 94 degrees of freedom

Multiple R-squared: 0.991, Adjusted R-squared: 0.9906

F-statistic: 2574 on 4 and 94 DF, p-value: < 2.2e-16

```

# conduct hypothesis test
linearHypothesis(model5, c("LNY70 = 1", "I(LNY70^2) = 0"))

```

Linear hypothesis test:

LNY70 = 1

I(LNY70^2) = 0

Model 1: restricted model

Model 2: LNC70 ~ LNY70 + I(LNY70^2) + LNP170 + LNP270

```

      Res.Df    RSS Df Sum of Sq      F      Pr(>F)
1         96 12.8296
2         94  1.7794   2      11.05 291.88 < 2.2e-16 ***

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

D.

I will now split the 1970 data into 5 subsamples, where the first 4 groups contain 20 rows and the last group contains 19. Next, I estimate by OLS the parameters of the part B. equation separately for each group. I find β_y coefficients for groups 1-5 of 0.675, 0.779, 1.141, 0.862, and 1.014 respectively. The corresponding returns to scale estimates are 1.481, 1.284, 0.876, 1.160, and 0.986 respectively. Relative to the 1955 data, the returns to scale estimates are much more moderate and seem to jump around a bit more. They also do not follow as consistent of a trend as we saw in Nerlove's estimation. In groups 1, 2, and 4 I see increasing returns to scale with r-values greater than 1. For groups 3 and 5 I see decreasing (but almost constant) returns to scale. Relative to Nerlove's findings, the 1970 returns to scale estimates I generated suggest a reduction in scale economies over time due to their smaller size relative to those in the 50's. The larger plants exhibit close to constant or slightly decreasing RTS generally with some exceptions, and RTS do not simply trend downwards in 1970 like was the case in the

50's. Instead of the U-shape described in earlier sections, these data take on more of an erratic shape with ups and downs as output increases. I think that this makes sense given what I understand about more modern mature industries, whose economies of scale may diminish but can experience expansionary bumps due to factors like technological advancement that allow them to push past physical limitations that may have been prohibiting.

```
# generate subsets of data
ss170 <- update_clean |>
  filter(
    Obs <= 20
  )
ss270 <- update_clean |>
  filter(
    Obs >20 & Obs <=40
  )
ss370 <- update_clean |>
  filter(
    Obs > 40 & Obs <=60
  )
ss470 <- update_clean |>
  filter(
    Obs>60 & Obs<= 80
  )
ss570 <- update_clean |>
  filter(
    Obs>80
  )

# run all models for individual subsets
model6 <- lm(data = ss170, LNC70~LNY70+LNP170+LNP270)
model7 <- lm(data = ss270, LNC70~LNY70+LNP170+LNP270)
model8 <- lm(data = ss370, LNC70~LNY70+LNP170+LNP270)
model9 <- lm(data = ss470, LNC70~LNY70+LNP170+LNP270)
model10 <- lm(data = ss570, LNC70~LNY70+LNP170+LNP270)
summary(model6)
```

Call:

```
lm(formula = LNC70 ~ LNY70 + LNP170 + LNP270, data = ss170)
```

Residuals:

Min	1Q	Median	3Q	Max
-----	----	--------	----	-----

-0.40755 -0.24735 0.05733 0.18731 0.41647

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-8.30757	1.44588	-5.746	3.01e-05 ***
LNy70	0.67502	0.04339	15.558	4.41e-11 ***
LNP170	0.33866	0.28322	1.196	0.249
LNP270	0.15756	0.21226	0.742	0.469

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2791 on 16 degrees of freedom

Multiple R-squared: 0.944, Adjusted R-squared: 0.9335

F-statistic: 89.96 on 3 and 16 DF, p-value: 3.132e-10

```
summary(model7)
```

Call:

```
lm(formula = LNC70 ~ LNy70 + LNP170 + LNP270, data = ss270)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.144232	-0.064771	-0.001432	0.056232	0.170654

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-9.81753	0.97976	-10.020	2.67e-08 ***
LNy70	0.77898	0.09283	8.391	2.97e-07 ***
LNP170	0.54720	0.15262	3.585	0.00247 **
LNP270	-0.15884	0.12571	-1.264	0.22448

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1002 on 16 degrees of freedom

Multiple R-squared: 0.8716, Adjusted R-squared: 0.8475

F-statistic: 36.2 on 3 and 16 DF, p-value: 2.321e-07

```
summary(model8)
```



```

Call:
lm(formula = LNC70 ~ LNY70 + LNP170 + LNP270, data = ss370)

Residuals:
      Min       1Q   Median       3Q      Max
-0.140904 -0.015412  0.008721  0.043109  0.135465

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -11.31902    0.98669  -11.47 3.94e-09 ***
LNY70         1.14123    0.09158   12.46 1.19e-09 ***
LNP170        0.25401    0.10121    2.51  0.0232  *
LNP270       -0.02767    0.13199   -0.21  0.8366
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08122 on 16 degrees of freedom
Multiple R-squared:  0.9271,    Adjusted R-squared:  0.9134
F-statistic: 67.81 on 3 and 16 DF,  p-value: 2.581e-09

```

```
summary(model9)
```

```

Call:
lm(formula = LNC70 ~ LNY70 + LNP170 + LNP270, data = ss470)

Residuals:
      Min       1Q   Median       3Q      Max
-0.13686 -0.05853 -0.01678  0.04257  0.24054

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -8.07959    1.78996  -4.514 0.000353 ***
LNY70         0.86206    0.15575   5.535 4.53e-05 ***
LNP170        0.12665    0.14841    0.853 0.406044
LNP270       -0.07009    0.12698   -0.552 0.588609
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09769 on 16 degrees of freedom
Multiple R-squared:  0.6949,    Adjusted R-squared:  0.6377
F-statistic: 12.15 on 3 and 16 DF,  p-value: 0.0002141

```

```
summary(model10)
```

Call:

```
lm(formula = LNC70 ~ LNY70 + LNP170 + LNP270, data = ss570)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.27966	-0.07606	0.01138	0.07105	0.24334

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-10.88381	1.16563	-9.337	1.22e-07	***
LNY70	1.01374	0.09048	11.205	1.10e-08	***
LNP170	0.36433	0.16038	2.272	0.0383	*
LNP270	-0.02695	0.17873	-0.151	0.8822	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1423 on 15 degrees of freedom

Multiple R-squared: 0.9019, Adjusted R-squared: 0.8823

F-statistic: 45.98 on 3 and 15 DF, p-value: 8.474e-08

```
# calculate returns to scale estimates for each
```

```
# ss170
```

```
1/0.675
```

```
[1] 1.481481
```

```
# ss270
```

```
1/0.779
```

```
[1] 1.283697
```

```
# ss370
```

```
1/1.141
```

```
[1] 0.8764242
```

```
# ss470  
1/0.862
```

```
[1] 1.160093
```

```
# ss570  
1/1.014
```

```
[1] 0.9861933
```

E.

The Christensen-Greene finding can be evaluated on several fronts, several of which I replicated in previous sections. By exploring for myself the returns to scale values for each subsection and comparing them to those found, by considering the context of the findings, and by relating them back to what makes sense intuitively about constraints, I was able to work through the 1070 findings and conclude in agreement with the finding that most U.S electricity was generated by firms who operated close to the min of their average cost curves. Comparing the 1970 results to the 1955 results, I find returns to scale estimates that are far closer to 1 (constant) implying that firms are seeing less in the way of large cost advantaged from further expansion. This is what we would expect of firms as they grow and increase production due to real world limitations like space, time, technology, and regulation. I also generated a plot similar to that in the first section showing residuals plotted against log of output by obs. As noted in part D. the curve is not as nicely U-shaped as the one generated using Nerlove's data, with most firms clustered around the flat bit where the residual value is approx 0 and the LNY70 is between 7 and 10 (on the higher end). This is the area that is directly before firms may start seeing negative returns to scale, which seems like a perfect place to stop pushing output increases. My estimates align with the Christensen-Greene finding, suggesting that by the 1970s, most of the economies of scale in the industry has been exploited and plants chose to operate relatively close to the efficient scale since producing less left cash on the table while producing more wouldnt reduce average cost for most.

```
# compute graphic showing residuals plotted against log of  
# output like in the first section to compare shape.  
update_clean <- update_clean |>  
  mutate(  
    resid = residuals(model5)  
  )  
  
# plot against log of output
```

```
ggplot(data = update_clean, mapping = aes(y = resid, x = LNY70, color = Obs)) +
  geom_point() +
  geom_smooth(method = loess) +
  theme_minimal() +
  labs(title = "Residuals Plotted Against Log of Output", x = "LNY70", y = "Residuals")
```

`geom\_smooth()` using formula = 'y ~ x'

Warning: The following aesthetics were dropped during statistical transformation:
colour.

- i This can happen when ggplot fails to infer the correct grouping structure in the data.
- i Did you forget to specify a `group` aesthetic or to convert a numerical variable into a factor?

